

TryHackMe - Autopsy

Autopsy is an open-source and powerful digital forensics platform.

Basic workflow:

1. Create/open the case for the data source you will investigate
2. Select the data source you wish to analyse
3. Configure the ingest modules to extract specific artefacts from the data source
4. Review the artefacts extracted by the ingest modules
5. Create the report

Note: Autopsy case files have a ".aut" file extension.

Data Sources

Autopsy can analyze multiple disk image formats.

Supported Disk Image Formats:

- **Raw Single** (For example: .img, .dd, .raw, .bin)
- **Raw Split** (For example: .001, .002, .aa, .ab, etc)
- **EnCase** (For example: .e01, .e02, etc)
- **Virtual Machines** (For example: .vmdk, .vhd)

Ingest Modules

Essentially **Ingest Modules** are Autopsy plug-ins. Each Ingest Module is designed to analyse and retrieve specific data from the drive. You can configure Autopsy to run specific modules during the source-adding stage or later by choosing the target data source available on the dashboard. By default, the Ingest Modules are configured to run on All Files, Directories, and Unallocated Space.

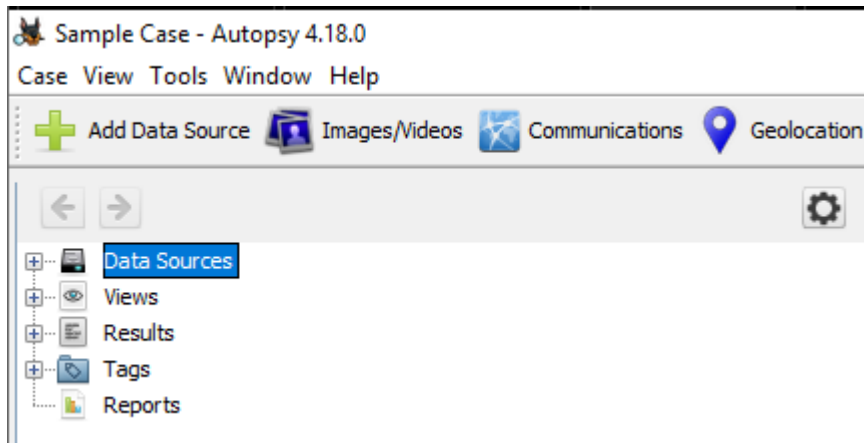
The User Interface

Let's look at the Autopsy user interface, which is comprised of 5 primary areas:

Tree Viewer

The **Tree Viewer** has **five top-level nodes**:

- **Data Sources** - all the data will be organised as you would typically see it in a normal Windows File Explorer.
- **Views** - files will be organised based on file types, MIME types, file size, etc.
- **Results** - as mentioned earlier, this is where the results from Ingest Modules will appear.
- **Tags** - will display files and/or results that have been tagged (read more about tagging [here](#)).
- **Reports** - will display reports either generated by modules or the analyst (read more about reporting [here](#)).



Result Viewer

Note: Don't confuse the Results node (from the Tree Viewer) with the Result Viewer.

When a volume, file, folder, etc., is selected from the Tree Viewer, additional information about the selected item is displayed in the Result Viewer.

Tip: When it comes to **File Types**, pay attention to this section. An adversary can rename a file with a misleading file extension. So the file will be 'miscategorised' **By Extension** but will be categorised appropriately by **MIME Type**

Contents Viewer

From the Table tab in the Result Viewer, if you click any folder/file, additional information is displayed in the Contents Viewer pane.

Listing					
/img_sample-case.dd/vol_vol3					
Table	Thumbnail	Summary			
Name	S	C	O	Modified Time	Change Time
\$BadClus:\$Bad				2015-03-25 11:08:36 GMT	2015-03-25 11:08:36 GMT
pagefile.sys				2015-03-25 13:05:36 GMT	2015-03-25 13:05:36 GMT
<					
Hex	Text	Application	File Metadata	Context	Results
Annotations					
Other Occurrences					
Name	/img_sample-case.dd/vol_vol3/pagefile.sys				
Type	File System				
MIME Type	application/octet-stream				
Size	2146951168				
File Name Allocation	Allocated				
Metadata Allocation	Allocated				
Modified	2015-03-25 13:05:36 GMT				
Accessed	2015-03-25 10:15:19 GMT				
Created	2015-03-25 10:15:19 GMT				
Changed	2015-03-25 13:05:36 GMT				
MD5	Not calculated				
SHA-256	Not calculated				
Hash Lookup Results	UNKNOWN				
Internal ID	211828				

In the given image, three columns might not be quickly understood what they represent.

- **S = Score**

The **Score** will show a red exclamation point for a folder/file marked/tagged as notable and a yellow triangle pointing downward for a folder/file marked/tagged as suspicious. These items can be marked/tagged by an Ingest Module or the analyst.

- **C = Comment**

If a yellow page is visible in the Comment column, it will indicate that there is a comment for the folder/file.

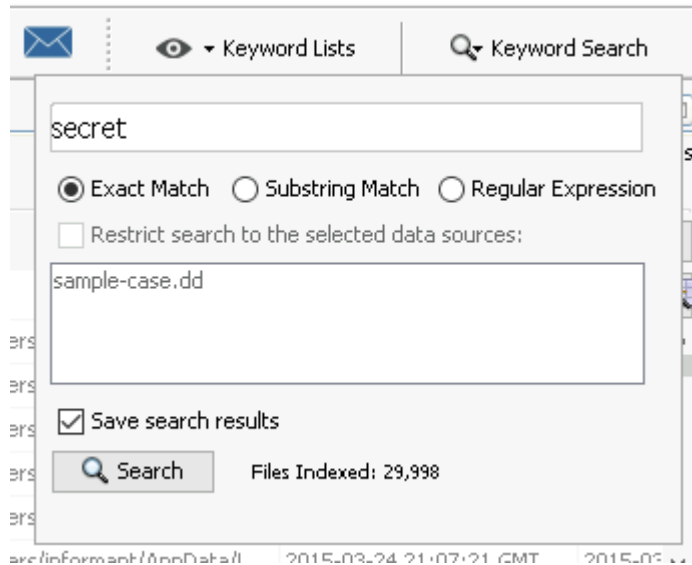
- **O = Occurrence**

In a nutshell, this column will indicate how many times this file/folder has been seen in past cases (this will require the [Central Repository](#).)

Refer to the Autopsy documentation on the Contents Viewer for more information [here](#).

Keyword Search

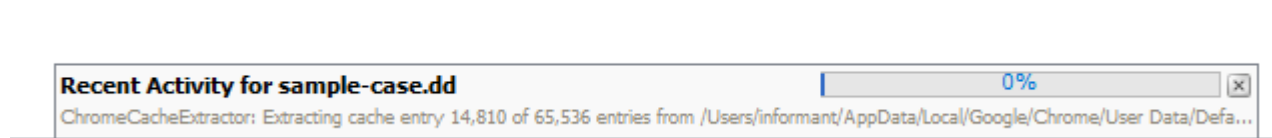
At the top right, you will find **Keyword Lists** and **Keyword Search**. With **Keyword Search**, an analyst can perform an AD-HOC keyword search.



Refer to the Autopsy [documentation](#) for more information on performing keyword searches with either option.

Status Area

Lastly, the **Status Area** is at the bottom right. When Ingest Modules run, a progress bar (along with the percentage completed) will be displayed in this area. More detailed information regarding the Ingest Modules is provided if you click on the bar.

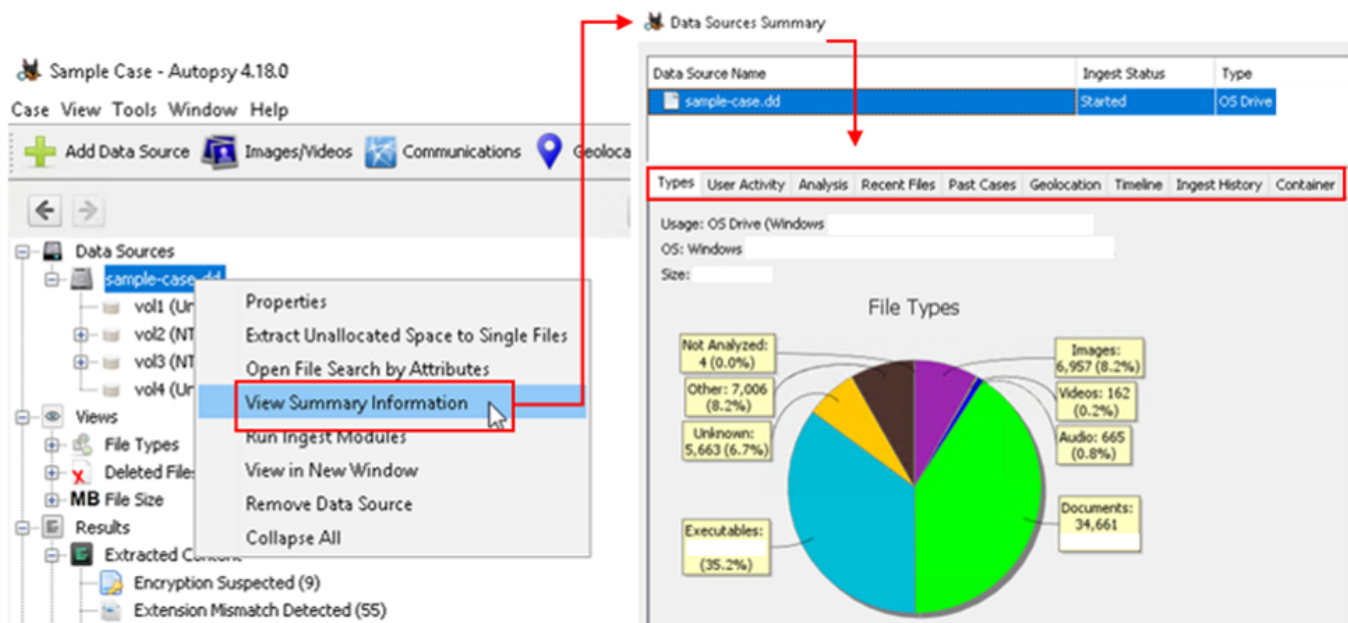


The User Interface II

Data Sources Summary

The **Data Sources Summary** provides summarized info in nine different categories. Note that this is an overview of the total findings. If you want to dive deep into the findings and look for a specific artefact, you need to analyze each module separately using the **"Result**

Viewer" shown in the previous task.

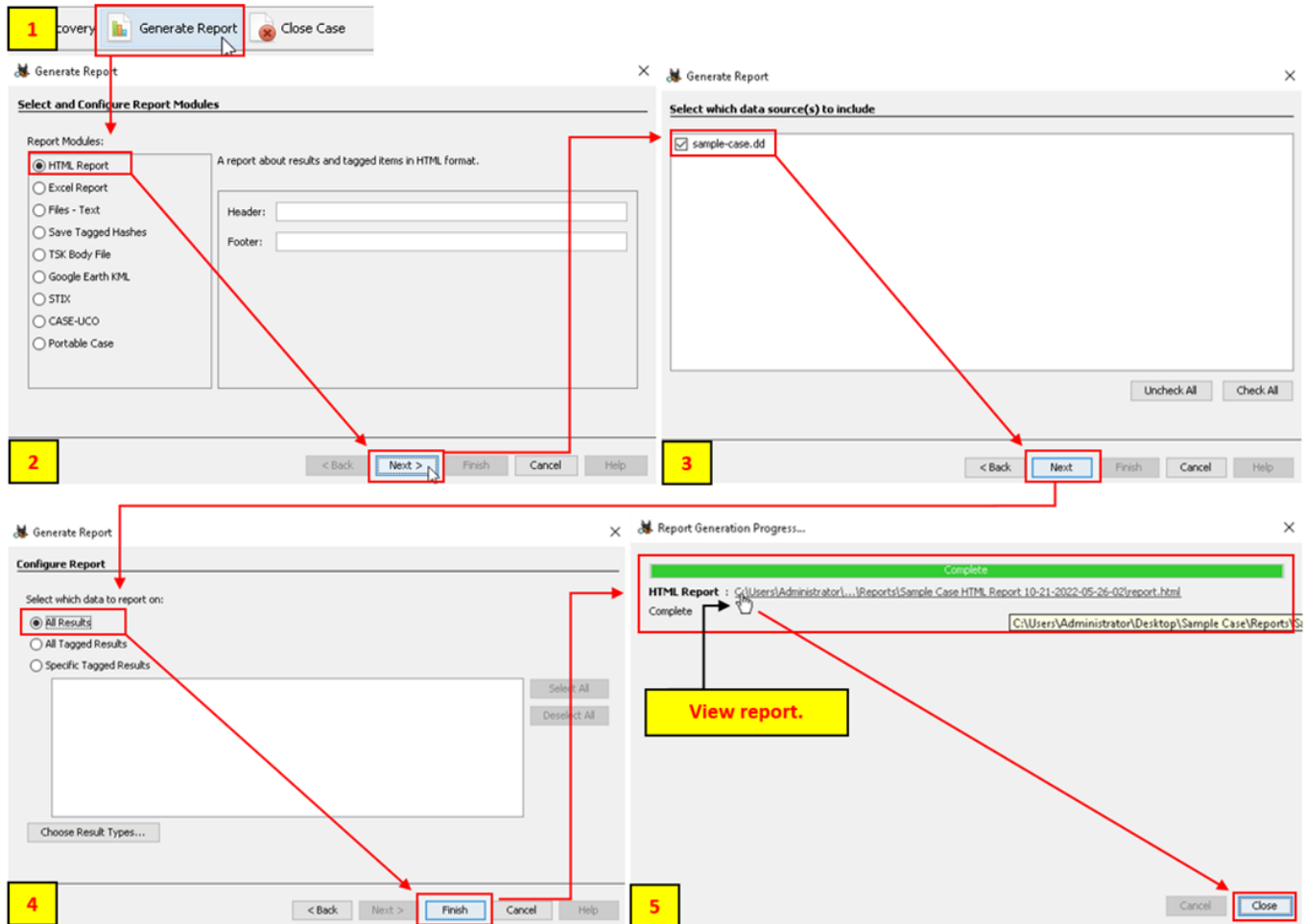


Generate Report

You can create a report of your findings in multiple formats, enabling you to create data sheets for your investigation case. The report provides all information listed under the "Result Viewer" pane. Reports can help you to re-investigate findings after finishing the live investigation. **However, reports don't have additional search options, so you must manually find artefacts for the event of interest.**

***Tip:** The Autopsy tool can be heavy for systems with low resources. Therefore completing an investigation with Autopsy on low resources can be slow and painful. Especially browsing long results might end up with a system freeze. You can avoid that situation by using reports. You can use the tool for parsing the data and generating the report, then continue to analyse through the generated report without a need for Autopsy. Note that it is always easier to conduct and manage an investigation with the GUI.*

You can use the **"Generate Report"** option to create reports. The steps are shown below.



Once you choose your report format and scope, Autopsy will generate the report. You can click on the "HTML Report" section (shown above) to view the report on your browser. Reports contain all of the "Result Viewer" pane results on the left side.